

NIKHIL KUMAR

AI/ML Engineer — Python — Machine Learning — DSA
Gaya, Bihar – 824206

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Professional Summary

AI/ML Engineer and Computer Science (Artificial Intelligence) graduate with hands-on experience in designing and developing end-to-end Machine Learning and AI-powered applications using Python, Flask, Scikit-learn, and Natural Language Processing (NLP). Strong foundation in Data Structures Algorithms, Database Management Systems, Operating Systems, Computer Networks, and Full-Stack Development. Experienced in building predictive analytics solutions, semantic NLP chatbots, academic intelligence systems, and responsive web dashboards. Passionate about leveraging AI and data-driven technologies to solve real-world problems, with a focus on scalable, efficient, and user-centric solutions.

Technical Skills

Programming: C, C++, Python, JavaScript
AI/ML: Pandas, NumPy, Scikit-learn, Deep Learning, CNN
Tools: Git, GitHub, VS Code
Core CS: Data Structures & Algorithms, OS, OOPs, DBMS, Computer Networks
Frontend: HTML, CSS, React.js, Tailwind CSS, Bootstrap, Responsive UI
Productivity Tools: Microsoft Excel, Microsoft Word, Microsoft PowerPoint

Experience

Smart City Intern (ICCC Systems) — [LINK](#) Dec 2025
Patna Smart City Limited (PSCL) Patna, Bihar

- Contributed to a 5-week ICCC internship supporting real-time Smart City monitoring systems including CCTV, GIS, ISWM, and alert management.
- Documented workflows for 3+ modules including ISWM, grievance handling, and pump automation monitoring to improve tracking efficiency.
- Prepared structured ICCC reports for dashboard monitoring and smooth operational shift handovers.

Projects

JARVIS – AI Voice Assistant [LINK](#) Dec 2025 – Jan 2026
AI voice assistant with speech recognition and automation

- Built an AI voice assistant to automate application launches, system commands, and web-based tasks using voice interaction.
- Integrated Speech-to-Text and Text-to-Speech modules to enable seamless hands-free control.

EduVision AI – Student Performance Prediction System [LINK](#) Feb 2026 – May 2026
AI-powered academic analytics platform for student performance prediction and risk detection

- Engineered a full-stack AI system to predict **student scores, grades, pass/fail status, and academic risk levels** using Random Forest, Logistic Regression, and Gradient Boosting models with **92% overall accuracy**.
- Designed responsive dashboards for **students and teachers** featuring bulk CSV/Excel prediction, analytics visualization, personalized AI mentor suggestions, and real-time academic monitoring using Flask, SQLite, Chart.js, and Scikit-learn.

House Price Prediction using Machine Learning [LINK](#) Jan 2026
End-to-end ML pipeline with model comparison and deployment

- Built a house price prediction system using the Ames dataset with preprocessing, feature engineering, and comparison of Linear Regression, Random Forest, and Gradient Boosting models.
- Selected Gradient Boosting as the best model ($R^2 = 0.91$) and deployed a Streamlit web app with a saved preprocessing + prediction pipeline.

Smart College Assistant [LINK](#) Jan 2025 – Feb 2025
AI-powered semantic NLP chatbot for smart college query assistance using Flask, Sentence Transformers, and Cosine Similarity.

- Implemented an intelligent chatbot with **semantic intent matching**, context-aware responses, smart follow-up suggestions, voice input support, and responsive dark/light UI for real-time student assistance.
- Implemented **Transformer-based NLP architecture** using Sentence Transformers and Scikit-learn to improve query understanding beyond keyword-based matching.

Education

B.Tech – Computer Science & Engineering (AI) 2022 – 2026
Bihar Engineering University, Patna CGPA: 8.51

Intermediate (10+2) – Science (PCM) 2019 – 2021
Mahavir College, Gaya (823003) – BSEB, Patna Percentage: 70.4%

Matriculation (Class 10) 2018 – 2019
Janta High School, Dumaria (824206) – BSEB, Patna Percentage: 67.2%

Certifications

- Completed industry-recognized certifications including **Python Essentials 2**, **Cybersecurity Essentials**, **Linux Unhatched (Cisco)**, **Problem Solving Through Programming in C (NPTEL — Top 5%)**, **Introduction to Operating Systems (NPTEL — Top 2%)**, and **IBM SkillBuild – Multimodal Retrieval-Augmented Generation (RAG)**.